

PolicyLens: Medical Policy

~~Intelligence~~ Platform

AI-assisted medical-policy review for a Blue Cross Blue Shield plan
(EBCBS)

The Executive Problem

- Medical-policy analysts review 200+ policies annually — by hand
- Reviews are disconnected from claims & utilization data
- Competitor policy comparisons are slow and inconsistent
- Financial impact of policy changes is guesswork

Review 200 policies in 1 week instead of 12, surface 3× more gaps, and simulate financial impact in real time — redirecting ~2,000 analyst hours per cycle to higher-value work.

The Status Quo — Manual, Fragmented, Slow

Annual policy review baseline — before PolicyLens

12
weeks

Annual review
cycle

~8 hrs

Per-policy
comparison

~2
wks

Actuarial sim
turnaround

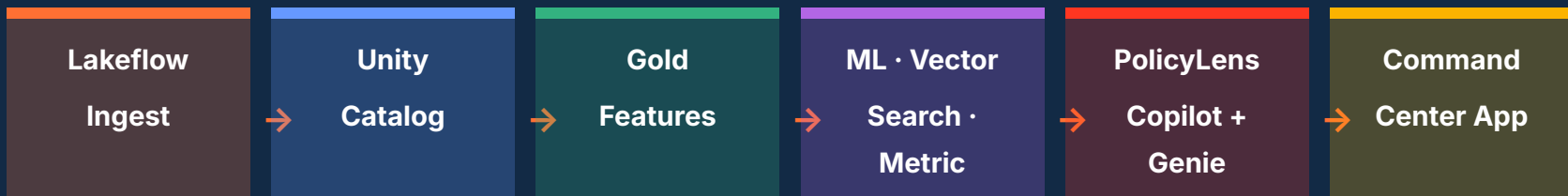
~15

Gaps found
per cycle

~2,400
hrs

Analyst hours
per cycle

What We Built — One Connected Data Journey



Lakebase serving the app · All six Databricks-native layers, integrated end-to-end

One platform, six layers:

- Lakeflow — declarative ingestion, Auto Loader, bronze→silver DQ expectations
- Unity Catalog — lineage, column masking on member PII, row-level security
- Gold Features — ML-ready curated tables
- ML · Vector Search · Metric View — LightGBM impact model, RAG, governed metrics
- PolicyLens Copilot (Claude / FMA) + Genie — agent loop + NL self-serve
- Command Center App — Databricks Apps, Lakebase-backed

Show — Data Foundation & Governance

Lakeflow Pipeline

~2M synthetic claims + medical policies,
prior-auth & utilization records ingested
via Declarative Pipeline

Auto Loader → Bronze → Silver

- Data-quality expectations on every layer
- Streaming-ready, incremental by default

3 realistic policy gaps seeded:

- Knee arthroscopy (overly permissive)

Spinal fusion (overly restrictive)

- Cardiac imaging (ambiguous language)

Unity Catalog Governance

End-to-end data lineage — column to
dashboard, automatically tracked

Column masking on member PII:

- member_id, dob, zip masked for
non-privileged analysts

Row-level security:

- Analysts see only their region's
claims unless VP+ role

Audit trail for all policy edits

and review-case state changes

Show — ML Intelligence

Financial-Impact Model (LightGBM)

Predicts annualized financial impact of
adding / removing CPT codes

$R^2 \approx 0.998$

MAE $\approx$ \$151K / month

Rule-Based Gap Detection

All 3 seeded gaps surfaced:

- Knee arthroscopy — overly permissive
(broad prior-auth approval pattern)
- Spinal fusion — overly restrictive

SHAP outputs explain which CPT-code clusters and prior-auth patterns drive each prediction — giving analysts the 'why' behind every financial-impact estimate. Budget-exposure context, not a recommendation to change coverage based on cost.

Show — PolicyLens Copilot Agent Loop

Governed Tools

- `compare_policies(p1, p2)`
- `get_utilization_data(cpt, region)`
- `analyze_prior_auth(policy_id)`
- `simulate_financial_impact(cpt_delta)`
- `get_industry_guidelines(cpt)`
- `create_review_case(policy_id, rec)`

PolicyLens Copilot Claude (FMA) + RAG + 6 Governed Tools

↓ opens review cases in Lakebase

Agent Scope & Guardrails

Scoped to policy admin
questions only

Declines individual
clinical / patient-care
questions

Never frames value as
cost savings — always

Compares EBCBS vs competitors · grounds in utilization/prior-auth/ML data · drafts recommendations · opens review cases in Lakebase

vector Search RAG over
policy document corpus

Show — Genie + The Command Center App

Genie — Natural Language Self-Serve

Governed Metric View powers NL queries
without raw table access

Example: "Show denial rate by region
for policy P-2024-CARD-01"

Result:

- Rochester: 9.8% denial rate
- Syracuse: 17.2%
- Albany: 24.6%
- Utica-Mohawk: 38.4% ← gap exposed

PolicyLens Command Center App

Databricks App + Lakebase backend

Modules:

- Executive KPI tiles
- Policy dashboard & search
- EBCBS vs. competitor comparison
- Real-time financial simulator
- Embedded Genie copilot
- Review-case management

Sub-second response — Lakebase

Ambiguous-criteria gap surfaces

automatically via variance signal

Impact — Before vs. After

KPI	Manual Baseline	With PolicyLens
Annual review cycle	~12 weeks	< 1 week
Time per policy comparison	~8 hours	~30 minutes
Financial simulation	~2 weeks	Real-time
Policy gaps per cycle	~15	~50
Analyst hours per cycle	~2,400	~400 (~2,000 redirected)

A 12-week manual cycle becomes a < 1-week governed, AI-assisted workflow.

Value for Both Personas

VP Medical Policy / CMO

Policy competitiveness

Stay current with competitor policies,
reviewed continuously not annually

Fewer appeals & admin overhead

Clear, defensible policy language
reduces ambiguity-driven denials

Admin efficiency

~2,000 analyst hours/cycle redirected
to clinical quality & member programs

Medical Policy Analyst Lead

Accurate comparisons in ~30 min

Structured side-by-side with
competitor data, not manual search

Real-time financial simulation

Instant impact estimate for any
CPT-code change, with SHAP context

Recommendations grounded in data

Plan's own claims, utilization &
prior-auth patterns, not anecdote

Better-informed decisions

Review cases auto-opened in Lakebase

How It Scales — A Reusable Governed Pattern

PolicyLens is built on a governed pattern: one lakehouse (Unity Catalog), Genie for self-serve, specialized agents composed over governed data. The same pattern extends across payer workflows.

Payment Integrity

Detect billing anomalies & overpayments using the same Gold features + ML model pattern.

Claims data already governed in Unity Catalog.

Appeals & Grievances

Agent-assisted case research: RAG over policy docs + member claims history. Analysts get a draft response in minutes, not days.

Provider Experience

Self-serve prior-auth status via Genie. Analysts answer provider queries with governed metrics, not manual lookup.

**"PolicyLens turns a 12-week manual cycle into
a governed, AI-assisted workflow:
better-informed policy decisions,
made faster, on one Databricks platform."**

All six Databricks layers built end-to-end:

Lakeflow · Unity Catalog · Lakebase · ML/Gen AI · Genie · Databricks Apps

All data synthetic · EBCBS is a code name · FE Bar submission

DATABRICKS · Field Engineering